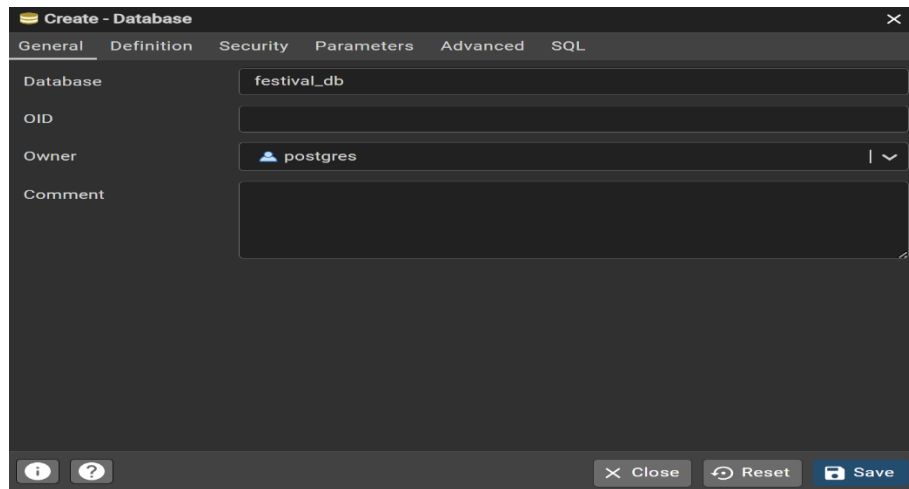


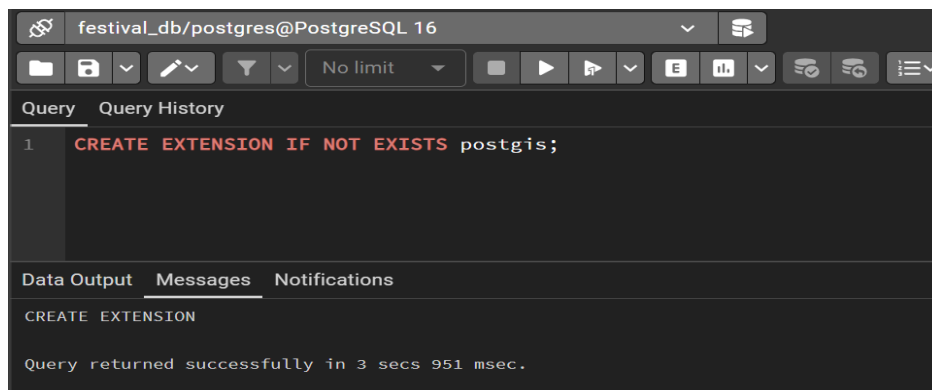
STEP BY STEP ANALYSIS OF THE PROJECT

The database was created "festival_db"

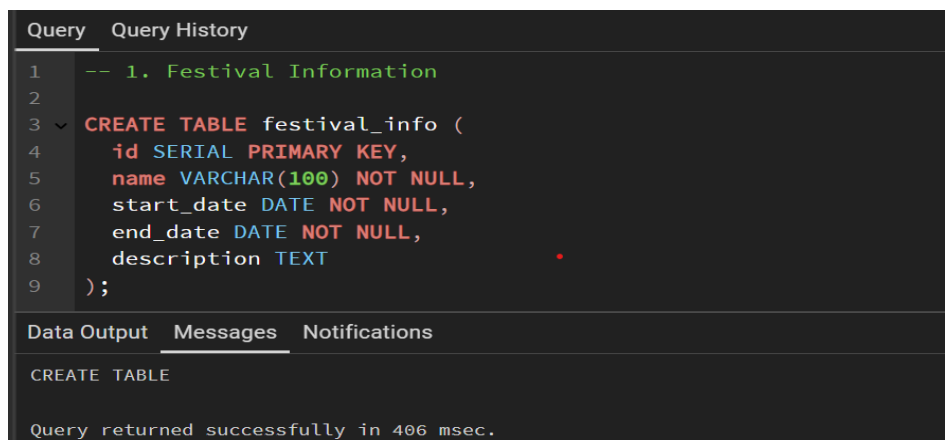
```
[CREATE DATABASE festival_db;]
```



PostGis extension was created on the database



Creating the festival information table



Inserting data into the festival information table

```
Query Query History
1  INSERT INTO festival_info (name, start_date, end_date, description)
2  VALUES
3  ('City Festival 2025', '2025-06-01', '2025-06-05',
4  'Annual city festival featuring events, attractions, and facilities.');
```

```
Data Output Messages Notifications
INSERT 0 1

Query returned successfully in 134 msec.
```

Creating the facility table

```
Query Query History
1  -- 2. Facility Table
2
3  CREATE TABLE facility (
4  id SERIAL PRIMARY KEY,
5  festival_id INT REFERENCES festival_info(id),
6  name VARCHAR(100) NOT NULL,
7  type VARCHAR(50) NOT NULL,
8  location GEOMETRY(Point, 4326) NOT NULL,
9  opening_time TIME,
10 closing_time TIME
11 );
```

```
Data Output Messages Notifications
CREATE TABLE

Query returned successfully in 478 msec.
```

Inserting data into the facility table

```
Query Query History
1  INSERT INTO facility (festival_id, name, type, location, opening_time, closing_time)
2  VALUES
3  (1, 'Main Toilet A', 'Toilet', ST_SetSRID(ST_MakePoint(13.0450, 47.8100), 4326), '08:00', '22:00'),
4  (1, 'Garbage Bin 1', 'Garbage Bin', ST_SetSRID(ST_MakePoint(13.0500, 47.8120), 4326), NULL, NULL),
5  (1, 'First Aid Station', 'First Aid', ST_SetSRID(ST_MakePoint(13.0480, 47.8110), 4326), '08:00', '22:00'),
6  (1, 'Information Booth', 'Info', ST_SetSRID(ST_MakePoint(13.0460, 47.8090), 4326), '09:00', '21:00'),
7  (1, 'Rest Area', 'Rest', ST_SetSRID(ST_MakePoint(13.0490, 47.8150), 4326), '10:00', '20:00'),
8  (1, 'Mobile Charging Station', 'Charging', ST_SetSRID(ST_MakePoint(13.0470, 47.8160), 4326), '10:00', '22:00'),
9  (1, 'Water Fountain', 'Water', ST_SetSRID(ST_MakePoint(13.0440, 47.8100), 4326), '08:00', '22:00'),
10 (1, 'Security Post', 'Security', ST_SetSRID(ST_MakePoint(13.0510, 47.8130), 4326), '07:00', '23:00'),
11 (1, 'Lost & Found', 'Info', ST_SetSRID(ST_MakePoint(13.0485, 47.8140), 4326), '09:00', '21:00'),
12 (1, 'Snack Kiosk', 'Kiosk', ST_SetSRID(ST_MakePoint(13.0455, 47.8120), 4326), '10:00', '20:00');
```

```
Data Output Messages Notifications
INSERT 0 10

Query returned successfully in 170 msec.
```

Creating the event's table

```
Query  Query History
1  -- 3. Event Table
2
3  CREATE TABLE event (
4      id SERIAL PRIMARY KEY,
5      festival_id INT REFERENCES festival_info(id),
6      name VARCHAR(100) NOT NULL,
7      description TEXT,
8      start_time TIMESTAMP NOT NULL,
9      end_time TIMESTAMP NOT NULL,
10     location GEOMETRY(Point, 4326) NOT NULL
11 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 126 msec.

Inserting data into the event's table

```
Query  Query History
1  INSERT INTO event (festival_id, name, description, start_time, end_time, location)
2  VALUES
3  (1, 'Opening Ceremony', 'Kick-off event for the festival', '2025-06-01 10:00', '2025-06-01 11:00', ST_SetSRID(ST_MakePoint(13.04, 42.33), 4326)),
4  (1, 'Concert in the Park', 'Live music performance', '2025-06-02 18:00', '2025-06-02 20:00', ST_SetSRID(ST_MakePoint(13.04, 42.33), 4326)),
5  (1, 'Art Exhibition', 'Local artists display their work', '2025-06-03 14:00', '2025-06-03 17:00', ST_SetSRID(ST_MakePoint(13.04, 42.33), 4326)),
6  (1, 'Street Performance', 'Theatrical and musical street acts', '2025-06-01 12:00', '2025-06-01 13:00', ST_SetSRID(ST_MakePoint(13.04, 42.33), 4326)),
7  (1, 'Food Tasting', 'Experience local cuisine', '2025-06-02 12:00', '2025-06-02 14:00', ST_SetSRID(ST_MakePoint(13.04, 42.33), 4326)),
8  (1, 'Dance Workshop', 'Interactive dance session', '2025-06-03 16:00', '2025-06-03 18:00', ST_SetSRID(ST_MakePoint(13.04, 42.33), 4326)),
9  (1, 'Fireworks Display', 'Spectacular end-of-day fireworks', '2025-06-05 21:00', '2025-06-05 21:30', ST_SetSRID(ST_MakePoint(13.04, 42.33), 4326)),
10 (1, 'Local Band Performance', 'Showcase of local talent', '2025-06-04 19:00', '2025-06-04 20:30', ST_SetSRID(ST_MakePoint(13.04, 42.33), 4326)),
11 (1, 'Cultural Parade', 'A festive parade through the city', '2025-06-04 15:00', '2025-06-04 16:30', ST_SetSRID(ST_MakePoint(13.04, 42.33), 4326)),
12 (1, 'Closing Ceremony', 'Festival wrap-up event', '2025-06-05 22:00', '2025-06-05 23:00', ST_SetSRID(ST_MakePoint(13.04, 42.33), 4326));
```

Data Output Messages Notifications

INSERT 0 10

Query returned successfully in 212 msec.

Creating the attraction's table

```
Query  Query History
1  -- 4. Attraction Table
2
3  CREATE TABLE attraction (
4      id SERIAL PRIMARY KEY,
5      festival_id INT REFERENCES festival_info(id),
6      name VARCHAR(100) NOT NULL,
7      type VARCHAR(50) NOT NULL,
8      geometry GEOMETRY,
9      opening_time TIME,
10     closing_time TIME
11 );
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 182 msec.

Inserting data into the attraction's table

```
Query Query History
1 INSERT INTO attraction (festival_id, name, type, geometry, opening_time, closing_time)
2 VALUES
3 (1, 'Food Booth 1', 'Booth', ST_SetSRID(ST_GeomFromText('POLYGON((13.0450 47.8100, 13.0460 47.8100, 13.0460 47.8110,
4 (1, 'Roller Coaster', 'Ride', ST_SetSRID(ST_MakePoint(13.0515, 47.8125), 4326), '10:00', '22:00'),
5 (1, 'Art Booth', 'Booth', ST_SetSRID(ST_GeomFromText('POLYGON((13.0445 47.8150, 13.0455 47.8150, 13.0455 47.8160, 13.
6 (1, 'Game Zone Ride', 'Ride', ST_SetSRID(ST_MakePoint(13.0475, 47.8140), 4326), '11:00', '21:00'),
7 (1, 'Tent Stage', 'Tent', ST_SetSRID(ST_GeomFromText('POLYGON((13.0480 47.8160, 13.0490 47.8160, 13.0490 47.8170, 13.
8 (1, 'VIP Lounge', 'Booth', ST_SetSRID(ST_GeomFromText('POLYGON((13.0465 47.8170, 13.0475 47.8170, 13.0475 47.8180, 13
9 (1, 'Ferris Wheel', 'Ride', ST_SetSRID(ST_MakePoint(13.0505, 47.8135), 4326), '10:00', '22:00'),
10 (1, 'Craft Market', 'Booth', ST_SetSRID(ST_GeomFromText('POLYGON((13.0440 47.8120, 13.0450 47.8120, 13.0450 47.8130,
11 (1, 'Kids Play Area', 'Attraction', ST_SetSRID(ST_GeomFromText('POLYGON((13.0435 47.8155, 13.0445 47.8155, 13.0445 47
12 (1, 'Interactive Exhibit', 'Exhibit', ST_SetSRID(ST_GeomFromText('POLYGON((13.0470 47.8180, 13.0480 47.8180, 13.0480

Data Output Messages Notifications
INSERT 0 10
Query returned successfully in 205 msec.
```

Creating the zone table

```
Query Query History
1 -- 5. Zone Table
2
3 CREATE TABLE zone (
4     id SERIAL PRIMARY KEY,
5     festival_id INT REFERENCES festival_info(id),
6     name VARCHAR(100) NOT NULL,
7     description TEXT,
8     area GEOMETRY(Polygon, 4326) NOT NULL
9 );

Data Output Messages Notifications
CREATE TABLE
Query returned successfully in 254 msec.
```

Inserting data into the zone's table

```
Query Query History
1 INSERT INTO zone (festival_id, name, description, area)
2 VALUES
3 (1, 'Food Area', 'Zone designated for food stalls and booths', ST_SetSRID(ST_GeomFromText('POLYGON((13.0440 47.8090,
4 (1, 'Entertainment Zone', 'Area for concerts and live shows', ST_SetSRID(ST_GeomFromText('POLYGON((13.0480 47.8100, 1
5 (1, 'Kids Zone', 'Designated safe play area for children', ST_SetSRID(ST_GeomFromText('POLYGON((13.0450 47.8150, 13.0
6 (1, 'Art Zone', 'Outdoor art installations and exhibitions', ST_SetSRID(ST_GeomFromText('POLYGON((13.0420 47.8120, 13
7 (1, 'VIP Zone', 'Exclusive area for VIP guests', ST_SetSRID(ST_GeomFromText('POLYGON((13.0500 47.8160, 13.0530 47.816
8 (1, 'Marketplace', 'Area with vendors and local crafts', ST_SetSRID(ST_GeomFromText('POLYGON((13.0460 47.8070, 13.048
9 (1, 'Cultural Zone', 'Exhibits showcasing local culture', ST_SetSRID(ST_GeomFromText('POLYGON((13.0480 47.8130, 13.05
10 (1, 'Game Zone', 'Area dedicated to interactive games and rides', ST_SetSRID(ST_GeomFromText('POLYGON((13.0430 47.815
11 (1, 'Relaxation Area', 'Quiet space for resting and socializing', ST_SetSRID(ST_GeomFromText('POLYGON((13.0400 47.816
12 (1, 'Main Stage Area', 'Primary performance area and gathering space', ST_SetSRID(ST_GeomFromText('POLYGON((13.0470 4

Data Output Messages Notifications
INSERT 0 10
Query returned successfully in 123 msec.
```

Creating spatial indexes to better optimize the performance

Query	Query History
1	-- 6. Spatial Indexes for Performance
2	
3	CREATE INDEX idx_facility_location ON facility USING GIST (location);
4	CREATE INDEX idx_event_location ON event USING GIST (location);
5	CREATE INDEX idx_attraction_geometry ON attraction USING GIST (geometry);
6	CREATE INDEX idx_zone_area ON zone USING GIST (area);

Data Output	Messages	Notifications
CREATE INDEX		
Query returned successfully in 128 msec.		

Creating a unified view called spatial entities

Query	Query History
1	-- 7. Unified View for Spatial Entities
2	
3	CREATE VIEW spatial_entities AS
4	SELECT id, name, 'Facility' AS entity_type, location AS geom
5	FROM facility
6	UNION ALL
7	SELECT id, name, 'Event' AS entity_type, location AS geom
8	FROM event
9	UNION ALL
10	SELECT id, name, 'Attraction' AS entity_type, geometry AS geom
11	FROM attraction
12	UNION ALL
13	SELECT id, name, 'Zone' AS entity_type, area AS geom
14	FROM zone;

Data Output	Messages	Notifications
CREATE VIEW		
Query returned successfully in 83 msec.		

SPATIAL QUERIES (USING DYNAMIC PARAMETERS)

Find Facilities within 500 Meters of a Given Location

Query

Query History

1

-- 1. Find Facilities within 500 Meters of a Given Location

2

SELECT id, name, type,

3

ST_Distance(location, ST_SetSRID(ST_MakePoint(13.045, 47.810), 4326)) AS distance

4

FROM facility

5

WHERE ST_DWithin(location, ST_SetSRID(ST_MakePoint(13.045, 47.810), 4326), 500)

6

ORDER BY distance;

Data Output

Messages

Notifications

<

Find Events within 500 Meters of a Given Location

Query

Query History

1

SELECT id, name, description, start_time, end_time,

2

ST_Distance(location, ST_SetSRID(ST_MakePoint(13.045, 47.810), 4326)) AS distance

3

FROM event

4

WHERE start_time > NOW() -- Only future events

5

AND ST_DWithin(location, ST_SetSRID(ST_MakePoint(13.045, 47.810), 4326), 500)

6

ORDER BY start_time;

Data Output

Messages

Notifications

SQL

	id	name	description	start_time	end_time	distance
	[PK] integer	character varying (100)	text	timestamp without time zone	timestamp without time zone	double precision
1	1	Opening Ceremony	Kick-off event for the festival	2025-06-01 10:00:00	2025-06-01 11:00:00	0.002236667977499345
2	4	Street Performance	Theatrical and musical street acts	2025-06-01 12:00:00	2025-06-01 13:00:00	0.005000000000000782
3	5	Food Tasting	Experience local cuisine	2025-06-02 12:00:00	2025-06-02 14:00:00	0.002236667977495373
4	2	Concert in the Park	Live music performance	2025-06-02 18:00:00	2025-06-02 20:00:00	0.004999999999999716
5	3	Art Exhibition	Local artists display their work	2025-06-03 14:00:00	2025-06-03 17:00:00	0.005099019513588217
6	6	Dance Workshop	Interactive dance session	2025-06-03 16:00:00	2025-06-03 18:00:00	0.006708203932496624
7	9	Cultural Parade	A festive parade through the city	2025-06-04 15:00:00	2025-06-04 16:30:00	0.0053851648071305295
8	8	Local Band Performance	Showcase of local talent	2025-06-04 19:00:00	2025-06-04 20:30:00	0.006999999999999787
9	7	Fireworks Display	Spectacular end-of-day fireworks	2025-06-05 21:00:00	2025-06-05 21:30:00	0.007211102550926938

Find Attractions within 1000 Meters of a Given Location

Query

Query History

1

2

3

4

5

SELECT id, name, type,

ST_Distance(geometry, ST_SetSRID(ST_MakePoint(13.045, 47.810), 4326)) AS distance

FROM attraction

WHERE ST_DWithin(geometry, ST_SetSRID(ST_MakePoint(13.045, 47.810), 4326), 1000)

ORDER BY distance;

Data Output

Messages

Notifications

<

Find the Nearest Facility to a Given Location

Query		Query History	
1	SELECT	id, name, type,	
2		ST_Distance(location, ST_SetSRID(ST_MakePoint(13.045, 47.810), 4326)) AS distance	
3	FROM	facility	
4	ORDER BY	location <-> ST_SetSRID(ST_MakePoint(13.045, 47.810), 4326)	
5	LIMIT	1;	
Data Output			
Messages			
Notifications			
SQL			
	id	name	type
	[PK] integer	character varying (100)	character varying (50)
			distance
			double precision
1	1	Main Toilet A	Toilet
			0

Find Facilities inside a Specific Zone

Query		Query History	
1	SELECT	f.id, f.name, f.type	
2	FROM	facility f	
3	JOIN	zone z ON ST_Contains(z.area, f.location)	
4	WHERE	z.name = 'Food Area';	
Data Output			
Messages			
Notifications			
SQL			
	id	name	type
	[PK] integer	character varying (100)	character varying (50)
1	1	Main Toilet A	Toilet
2	10	Snack Kiosk	Kiosk

Find the Total Number of Events in Each Zone

Query		Query History	
1	SELECT	z.name AS zone_name, COUNT(e.id) AS event_count	
2	FROM	zone z	
3	LEFT JOIN	event e ON ST_Within(e.location, z.area)	
4	GROUP BY	z.name	
5	ORDER BY	event_count DESC;	
Data Output			
Messages			
Notifications			
SQL			
	zone_name	event_count	
	character varying (100)	bigint	
1	Entertainment Zone	2	
2	Food Area	1	
3	Game Zone	1	
4	Art Zone	1	
5	Marketplace	0	
6	Relaxation Area	0	
7	VIP Zone	0	
8	Cultural Zone	0	
9	Kids Zone	0	
10	Main Stage Area	0	
Total rows: 10 of 10			
Query complete 00:00:00.353			
Ln 5, Col 27			

Find the Number of Facilities within Each Zone

Query		Query History	
1	SELECT	z.name AS zone_name, COUNT(f.id) AS facility_count	
2	FROM	zone z	
3	LEFT JOIN	facility f ON ST_Within(f.location, z.area)	
4	GROUP BY	z.name	
5	ORDER BY	facility_count DESC;	

Data Output		Messages		Notifications	
	zone_name		facility_count		
	character varying (100)		bigint		
1	Food Area		2		
2	Cultural Zone		2		
3	Kids Zone		1		
4	Main Stage Area		1		
5	Entertainment Zone		1		
6	VIP Zone		0		
7	Game Zone		0		
8	Marketplace		0		
9	Relaxation Area		0		
10	Art Zone		0		
Total rows: 10 of 10 Query complete 00:00:00.861 Ln 5, Col 30					

Find Security-Related Facilities (Security Posts, First Aid, Police Stations)

Query		Query History	
1	SELECT	id, name, type, location	
2	FROM	facility	
3	WHERE	type IN ('Security', 'First Aid', 'Info');	

Data Output		Messages		Notifications	
	id	name	type	location	
	[PK] integer	character varying (100)	character varying (50)	geometry	
1	3	First Aid Station	First Aid	0101000020E61000007F6ABC7493182A402B8716D9CEE747...	
2	4	Information Booth	Info	0101000020E6100000643BDF4F8D172A40643BDF4F8DE747...	
3	8	Security Post	Security	0101000020E6100000273108AC1C1A2A40F2D24D6210E847...	
4	9	Lost & Found	Info	0101000020E610000046B6F3FDD4182A40D578E92631E84740	

Find the distance between two facilities

Query		Query History	
1	SELECT	f1.name AS facility_1, f2.name AS facility_2,	
2		ST_Distance(f1.location, f2.location) AS distance_meters	
3	FROM	facility f1, facility f2	
4	WHERE	f1.name = 'First Aid Station' AND f2.name = 'Main Toilet A';	

Data Output		Messages		Notifications	
	facility_1	facility_2	distance_meters		
	character varying (100)	character varying (100)	double precision		
1	First Aid Station	Main Toilet A	0.00316227766016775		

Find all attractions and events within 500 meters of a given point

Query

Query History

1

SELECT 'Event' AS category, id, name, start_time, end_time,

2

ST_Distance(location, ST_SetSRID(ST_MakePoint(13.045, 47.810), 4326)) AS distance

3

FROM event

4

WHERE start_time > NOW()

5

AND ST_DWithin(location, ST_SetSRID(ST_MakePoint(13.045, 47.810), 4326), 500)

6

UNION ALL

7

SELECT 'Attraction' AS category, id, name, NULL AS start_time, NULL AS end_time,

8

ST_Distance(geometry, ST_SetSRID(ST_MakePoint(13.045, 47.810), 4326)) AS distance

9

FROM attraction

10

WHERE ST_DWithin(geometry, ST_SetSRID(ST_MakePoint(13.045, 47.810), 4326), 500)

11

ORDER BY distance;

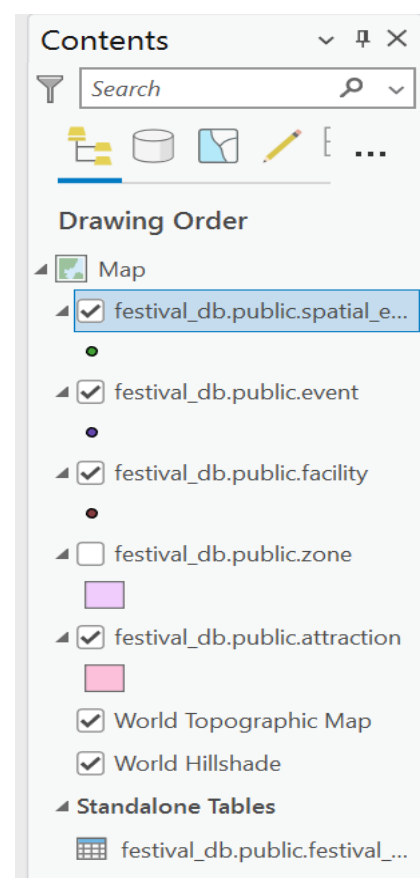
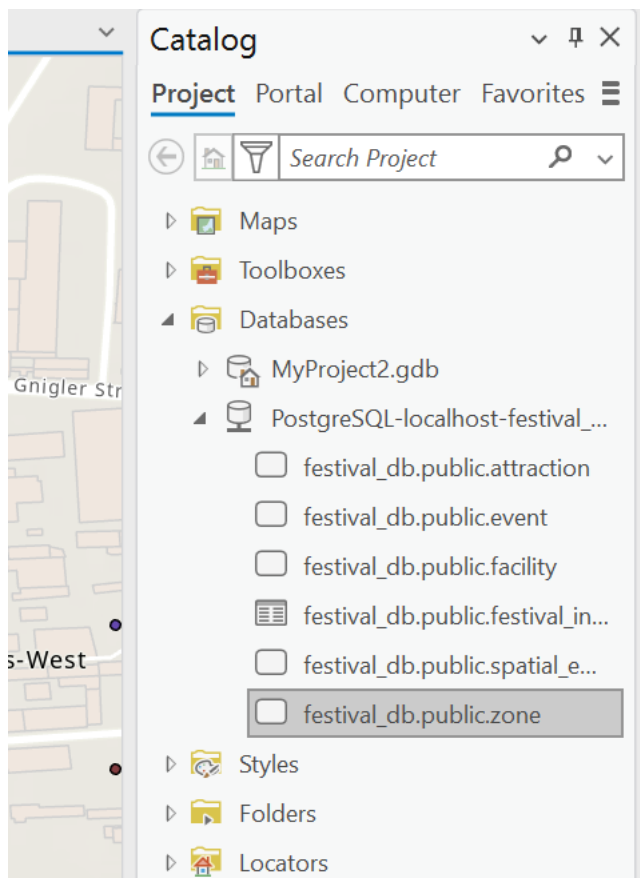
Data Output

Messages

Notifications

</

ADDING A DATABASE CONNECTION IN ARCGIS PRO



ADDING A DATABASE CONNECTION IN QGIS

